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Q1. Harshad number

ANSWER

```
def is_harshad(n):
    s=0
    temp=n
    while temp>0:
        s+=temp%10
        temp//=10
    return n%s==0
n=int(input("Enter a number"))
if is_harshad(n):
    print(n,"is a harshad number")
else:
    print(n, "is not a harshad number")
print("harshad numbers between 1 and 500")
for i in range (1,501):
    if is_harshad(i):
        print(i,end=" ")
```

Q2. GCD

CODE

```
a=int(input("enter first:"))
b=int(input("enter second:"))
while b!=0:
    temp=a
    a=b
    b=temp%b
print("GCD is",a)
```

INPUT

```
12
18
```

OUTPUT

```
enter first:enter second:GCD is 18
GCD is 12
GCD is 6
```

Q3. sum to a target value

ANSWER

```
set()
lst
num
seen
seen
pairs
find_pairs
```

Q4. Bitwise Operators

ANSWER

```
a=0b1101=13
b=0o25=21
c=0x1a=26
result=(a&b)|c
        =(13&21)|26
        =5|26
        =31
```

Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[mid]==target:
            return mid
        if arr[low]<=arr[mid]:
            if target<=arr[low]and target< arr[mid]:
                high=mid-1
            else:
                low=mid+1
        else:
            if target>arr[mid]and target<=arr[high]:
                low=mid+1
            else:high=mid+1
            high=mid-1
            return-1
    data=[4,5,6,7,0,1,2]
    t=int(input("target:"))
    result=rotated_binary_search(data,t)
    print("found at:"result)
```

OUTPUT

```
File "C:\Windows\TEMP\sandbox_192419061RC_6a625f66d64282.73809885\main.py", line 13
    else:
    ^^^^^
SyntaxError: invalid syntax
```